

Sarah-Sonia Balan

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Education

2023-2027 **The University of Manchester**
MMath Mathematics

- Year averages: 84.5% (Year 1), 80.2% (Year 2), and 80.0% (Year 3); ranked in the top 10% of the cohort in Years 2 and 3 (departmental data).
- Selected results: Algebraic Geometry 88, Number Theory 86, Double Project 86, Advanced Algebra 81, Mathematics Education 80.
- Final-year modules include Category Theory, Representation Theory, Galois Theory, Algebraic Number Theory, Lie Algebras, and Algebraic Topology.
- MMath project (from September 2026), supervised by Dr Martin Orr: proposed direction on supersingular elliptic curves and quaternionic endomorphism algebras.

Research & Academic Experience

Jun – Aug 2026 **Count Me In Research Internship, Algebra–Geometry–Quantum CDT**,
University of Edinburgh

“Quantum knot invariants built from typical representations of the $U_h(\mathfrak{sl}(3|1))$ are q-holonomic”,
supervisor: Dr Jennifer Brown

- Selected as one of nine students from approximately 200 applicants for a competitive six-week research internship.
- Led the algebraic and representation-theoretic strand of a three-person project; the collaboration continues beyond the funded internship.
- Developed a PBW basis description of the typical induced module for $U_h(\mathfrak{sl}(3|1))$ to organise its odd basis vectors and predict their weights.
- Connected the construction with embedded $U_h(\mathfrak{sl}(2|1))$ subalgebras and began investigating its extension into a general framework for $U_h(\mathfrak{sl}(n|1))$.

Sep 2025 – May 2026 **Noetherianity of Noncommutative Polynomial Rings —**
Individual Double Project, *University of Manchester*

Supervisor: Dr Omar León Sánchez; mark: 86% (one of the five highest-scoring double projects in the cohort)

- Synthesised extensions of Hilbert’s Basis Theorem from automorphism-twisted polynomial rings, formal differential operator rings, and general Ore extensions.
- Presented formal differential operator rings through an endomorphism ring construction and analysed the recovery of prime ideals from invariant ideals in the base ring.
- Undertook self-directed study from Hungerford and Goodearl–Warfield, and designed the report’s historical framing and exposition.

Jun - Sep 2025 **Facilitating Effective Learning in Abstract Algebra: A Study of Teaching Practices and Proof Comprehension — LMS Undergraduate Research Bursary, University of Manchester**

Mathematics Education research, supervisor: Professor Louise Walker

- Combined a literature review with six independently conducted and analysed semi-structured interviews to identify barriers to learning proof and abstraction.
- Co-developed evidence-informed activities using concept images and example spaces, inquiry-based learning, concept tests, active self-feedback, and self-explanation.
- Produced the final report and teaching toolkit; the project contributed to a co-authored MSOR Connections article and several conference presentations.

Publications and Mathematical Writing

- Aug 2026 Walker, L., Marshall, S., and Balan, S.-S. "1+1 is greater than 1: Developing Scholarship with Students as Partners." *MSOR Connections* 24(2), 5–10.
- Apr 2026 Balan, S. "Significant Figures: Emmy Noether." *Chalkdust*, Issue 23.
- Dec 2025 Balan, S.-S. "The Summer I Became a Mathematician: Reflections on the LMS Summer School and My Project." *LMS Newsletter*, 517.
- Jan 2025 Balan, S. "The big journey to Fermat's little theorem." *The Mancunion*.

Selected Presentations

- Sep 2026 **"Stripping Away the Inessentials: Emmy Noether & the Rise of Noncommutative Algebra."** Piscopia PiForum26 Conference.
- Sep 2026 **Co-presenter on students-as-partners work.** RAISE Conference; with Professor Louise Walker.
- Jun 2026 **Co-keynote speaker, Students as Partners in the Mathematical Sciences.** UK-wide workshop organised with the London Mathematical Society and Institute of Mathematics and its Applications; with Professor Louise Walker.
- Apr 2026 **"Meaningful dialogue and flexibility — student voices and staff perspectives on a co-creation approach to advising."** Lightning talk, UKAT Annual Conference; with Dr Neil Morrison.
- Mar 2026 **"Emmy Noether's Ascending Chain of Abstraction."** Manchester Interdisciplinary Mathematics Undergraduate Conference (MIMUC), University of Manchester. First Prize.
- Feb 2026 **"Emmy Noether: Life and Mathematics."** Poster pitch and poster, Oxford Women and Non-Binary People in Mathematics Day (OxWIM Day), University of Oxford.
- Jul 2025 **"Generating and Communicating Ideas: Co-creation and Student Voice in Teaching Materials."** Poster, University of Manchester Institute of Teaching and Learning Conference; with the Library Student Team.
- Jul 2025 **"Classic UoM — Student Experience Issues."** Lightning talk, University of Manchester Institute of Teaching and Learning Conference; with Kai Prince and Dr Neil Morrison.

Teaching and Mathematics Education Experience

2025–2026 **PASS (Peer-Assisted Study Sessions) Coordinator, Department of Mathematics**, University of Manchester

Designed weekly resources across the first-year mathematics curriculum, trained 43 PASS Leaders, and used observation and structured feedback to refine materials.

2024–2025 **PASS Leader, Department of Mathematics**, University of Manchester

2023–2025 **Mathematics tutor, Levelling Up in Mathematics and mA*ths**. Supported learners from groups under-represented in higher education with A-level mathematics and university preparation.

2020–2023 **Private mathematics tutor, Romania**. Individual tuition primarily for learners aged around 14.

Academic Leadership and Service

2023–2026 **Student Development Secretary, University of Manchester MathSoc**. Revived and co-organised the Manchester Maths League, including an edition attracting more than 70 participants; coordinated problem-solving sessions and accessible research talks.

2025–2026 **Core student organiser, Manchester Interdisciplinary Mathematics Undergraduate Conference**. Contributed to speaker outreach, sponsorship, publicity, and logistics for a two-day conference attended by more than 200 participants in 2026.

2025–present **Student Representative Coordinator, Department of Mathematics**. Supported student representatives and contributed student perspectives to department-level discussions and improvement work.

2024–present **Library Student Team — Training and Development / Quality Assurance Area Lead**. Reviewed and co-created inclusive learning resources, mentored student team members, and supported study-skills workshops and student-partnership work.

Awards, Funding and Professional Recognition

Mar 2026 **Fellow of the Higher Education Academy (FHEA)**, Advance HE.

Mar 2026 **Faculty of Science and Engineering Undergraduate of the Year** — departmental nominee.

Mar 2026 **First Prize, MIMUC**.

Jul 2025 **London Mathematical Society Undergraduate Summer School, University of Bath**. One of 50 students selected nationally for the residential programme.

Jun 2025 **London Mathematical Society Undergraduate Research Bursary**. Co-funded by the University of Manchester.

May 2025 **Third Place, Volunteer of the Year — Student Supporting Other Students**, University of Manchester.

Additional Skills

Technical skills: LaTeX and Python, with working familiarity in MATLAB, C++, SQL

Languages: Romanian (native), English (C2, certified), German (B1, certified), Polish (A2)